Observational Hydrology in Data-Sparse Catchments: Part A — Empirical Hydro-Geomorphic Inundation Modeling in the Lower Niger-Benue Basin, Nigeria.

ABSTRACT

Hydrodynamic modeling in sub-Saharan Africa is severely constrained by a chronic lack of in-situ gauge telemetry. To benchmark 2D hydrodynamic simulations in ungauged transboundary catchments, physical hydrologists require empirical boundary conditions. This study engineers a cloud-native Earth Observation pipeline utilizing multi-temporal Sentinel-1 Synthetic Aperture Radar (SAR) and MERIT-Hydro topographic data to extract flood inundation frequencies across the Lower Niger-Benue confluence. By applying an adaptive backscatter threshold coupled with a deterministic terrain slope constraint, the pipeline bypassed persistent monsoon cloud occlusion. Validation against the JRC Global Surface Water dataset yielded a **99.36% baseline recall**, while simultaneously discovering over 2,497 km² of hidden peak-monsoon inundation missed by optical sensors.

1. INTRODUCTION

A critical bottleneck in tropical hydrology is the gap between spaceborne measurement and downstream engineering interventions. In vast transboundary systems like the Lower Niger and Benue River catchments, physical hydrologists frequently lack the empirical boundary parameters (e.g., historical flood extents and channel retention capacities) required to run deterministic routing models such as HEC-RAS or LISFLOOD.

Without these empirical states, early warning architectures fail. This project was developed to overcome this observational deficit by engineering an automated, reproducible workflow in Python and the Google Earth Engine (GEE) API. The objective is to extract historical flood-pulse dynamics from the active microwave spectrum, mathematically couple them with topographic surfaces, and derive a Catchment Vulnerability Index (CVI) to inform physical flood defense strategies in data-sparse regions.

The spatial domain for this research encompasses the Lower Niger-Benue confluence, centered at Lokoja, Nigeria (Figure 1.0). This critical junction merges two of West Africa's largest transboundary river networks. The region is characterized by a complex, low-relief floodplain that experiences severe seasonal volumetric expansion driven by the West African Monsoon, making it a vital, yet notoriously data-sparse, corridor for hydrodynamic modeling.

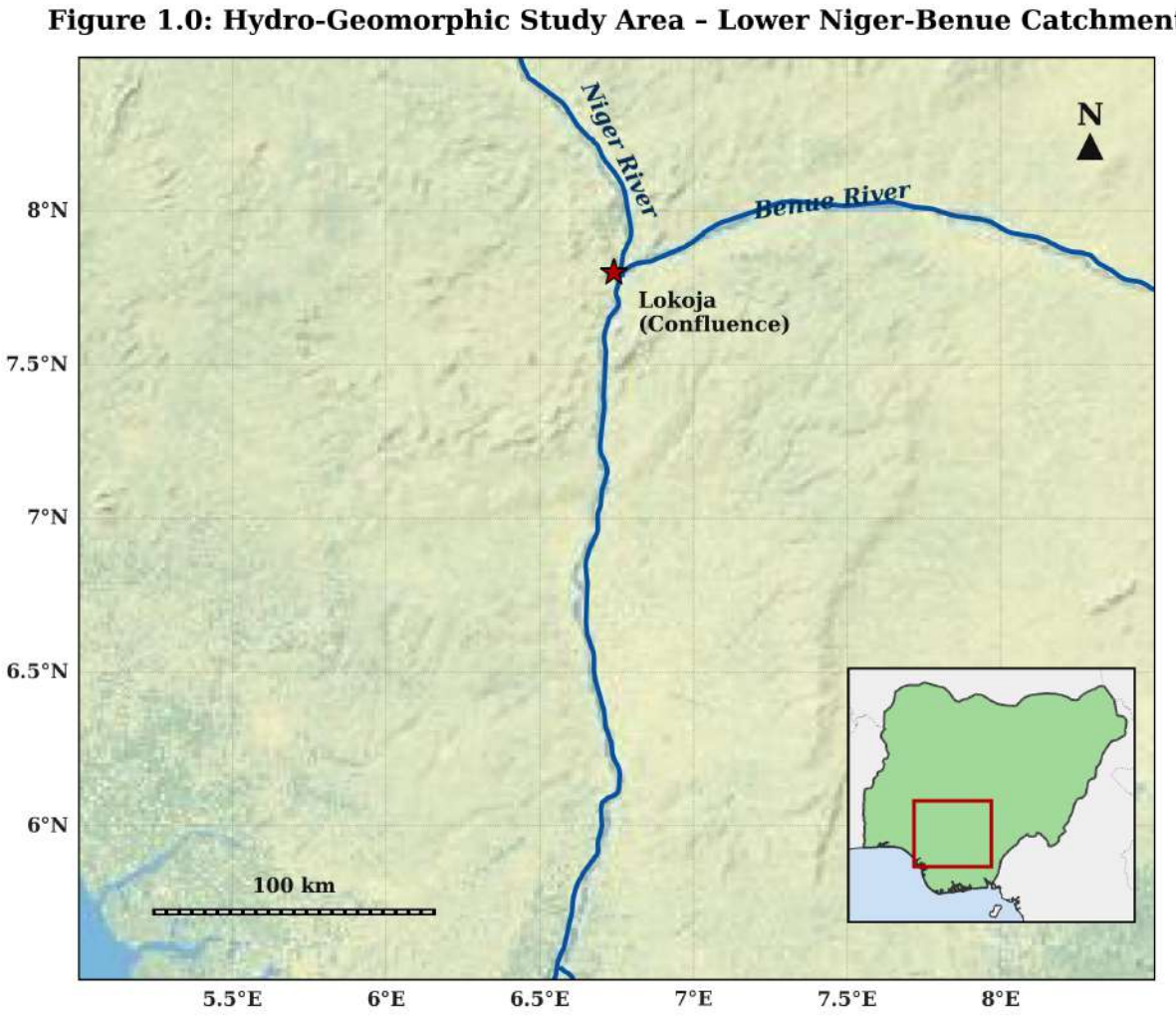


Figure 1.0: Hydro-Geomorphic Study Area – Lower Niger-Benue Catchment

Because this specific catchment is heavily obscured by dense monsoon cloud cover during peak flood months, traditional optical satellites are ineffective for continuous inundation mapping. To overcome this environmental constraint, the analytical framework transitions from passive observation to active microwave telemetry, enabling clear, all-weather detection of the basin's hydrological extremes.

2. DATA AND METHODOLOGY

The computational architecture relies on a hybrid cloud-to-local Python pipeline to process multi-terabyte satellite arrays, avoiding the hardware limitations of traditional desktop GIS.

2.1 Cloud-Native SAR Physics (Sentinel-1)

To bypass the persistent Sahelian-Guinean monsoon cloud cover, active microwave telemetry from Copernicus Sentinel-1 (Interferometric Wide Swath) was utilized. Dual-polarized backscatter (γ^0) arrays were filtered using a focal median spatial reducer to suppress radar speckle. A baseline pre-monsoon state (Jan-Mar) was isolated and compared against the extreme flood-pulse state (Sep-Nov) to map the fluvial boundary expansion.

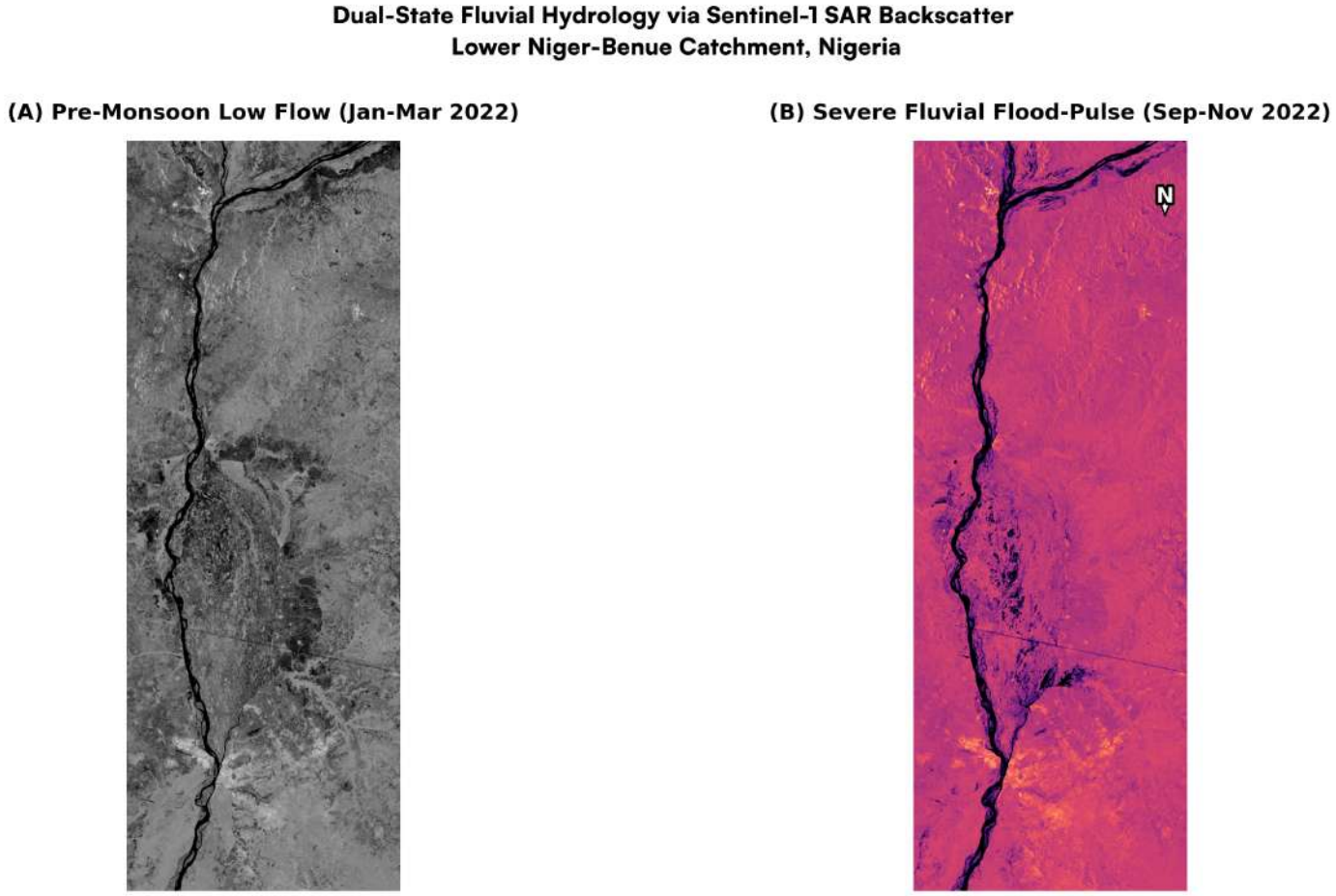


Figure 2: Dual-State Fluvial Hydrology (Baseline vs Peak Flood-Pulse)

2.2 Multi-Temporal Inundation Frequency & Topographic Conditioning

Because mapping a single historical event is insufficient for engineering design, a 5-year longitudinal analysis (2018–2023) was executed to determine the spatial recurrence of inundation. To mathematically isolate standing water, a physics-based threshold was applied: VH polarization backscatter < -18 dB.

Crucially, to eliminate the false-positives common in global radar thresholding (caused by radar shadow in complex topography), a geomorphic mask was introduced. The 30m MERIT-Hydro Digital Elevation Model (DEM) was ingested to calculate continuous terrain slope. By masking all pixels with a slope > 5°, the inundation algorithm was strictly confined to physical pooling zones and active floodplains.

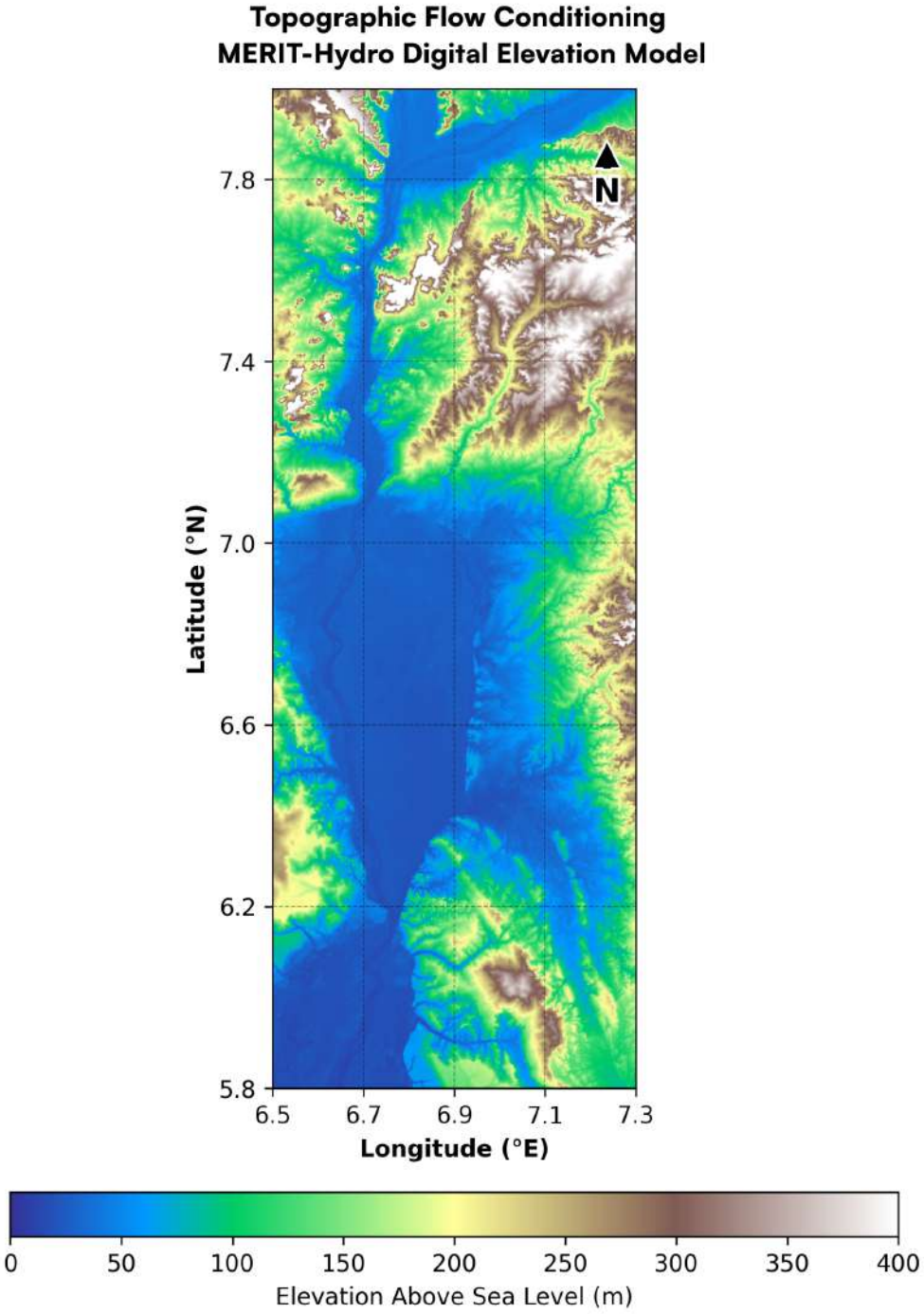


Figure 3: Topographic Flow Conditioning (MERIT-Hydro DEM)

2.3 Hydro-Geomorphic Vulnerability Modeling

The Catchment Vulnerability Index (CVI) was derived via a Multi-Criteria Evaluation (MCE) computed in local Python memory (NumPy). The algorithm weights the multi-year inundation frequency against the

normalized inverse elevation of the MERIT-Hydro DEM, highlighting low-lying communities subjected to high-frequency fluvial extremes.

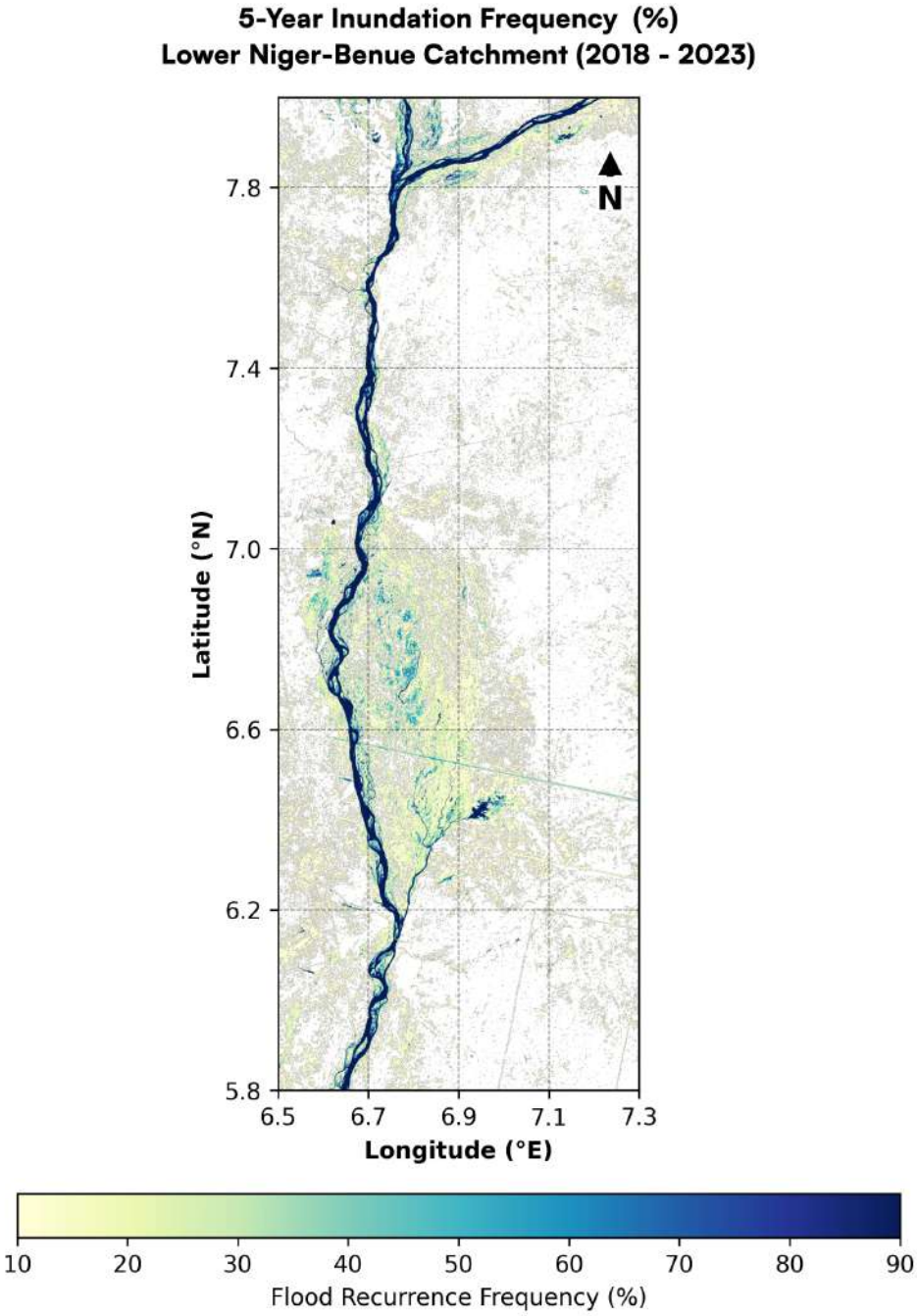


Figure 4: Five-Year Inundation Frequency Surface (%)

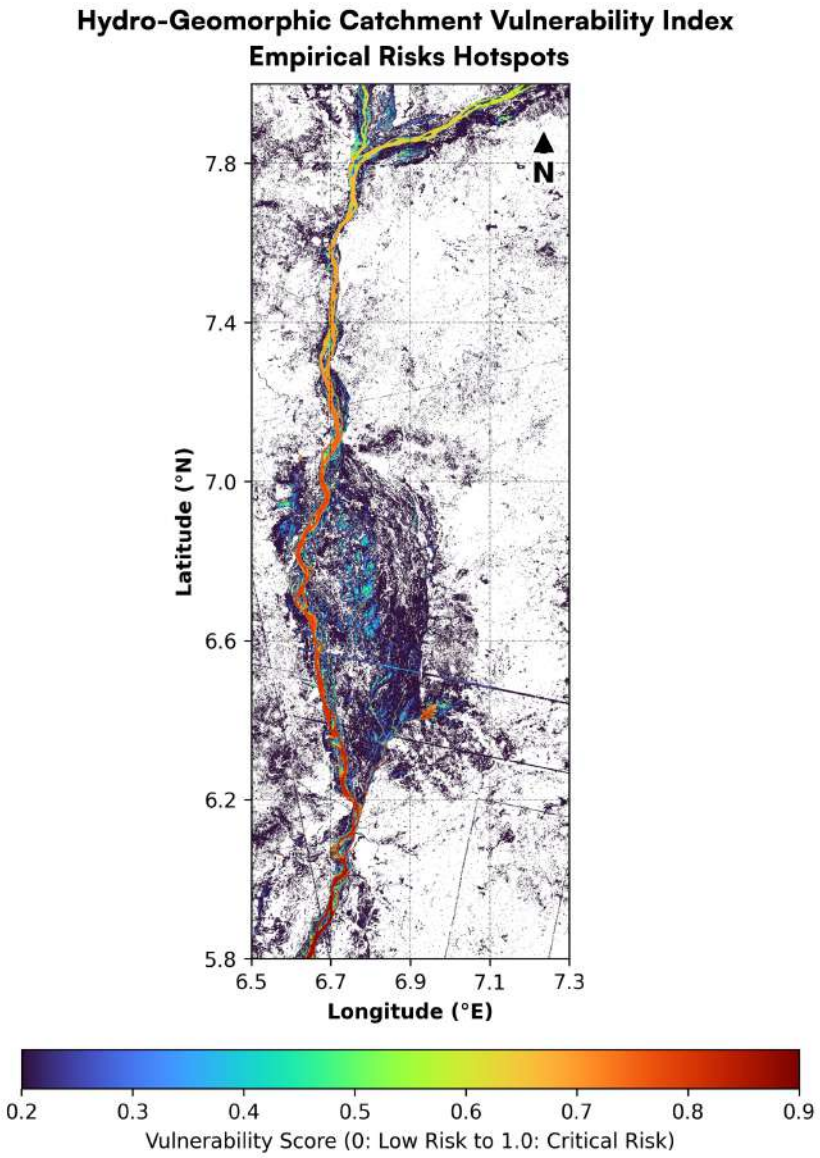


Figure 5: Hydro-Geomorphic Catchment Vulnerability Hotspots

3. RESULTS AND DISCUSSION

The pipeline successfully extracted quantitative hydrological metrics from a largely ungauged basin.

3.1 Seasonal Hydro-Dynamics

The synthesized seasonal hydro-period curve quantitatively tracks the flood pulse, demonstrating a sharp volumetric expansion from 45 km² of inundated area in the dry season to a peak extreme of 1,400 km² by

October. Alongside the 1D hydraulic cross-section, this empirical curve provides the precise temporal boundary constraints required by hydrologists to calibrate simulated discharge routing.

Furthermore, the MCE Vulnerability Hotspot map translates these satellite physics into an actionable engineering deliverable, delineating critical risk corridors downstream toward the Anambra basin sinks.

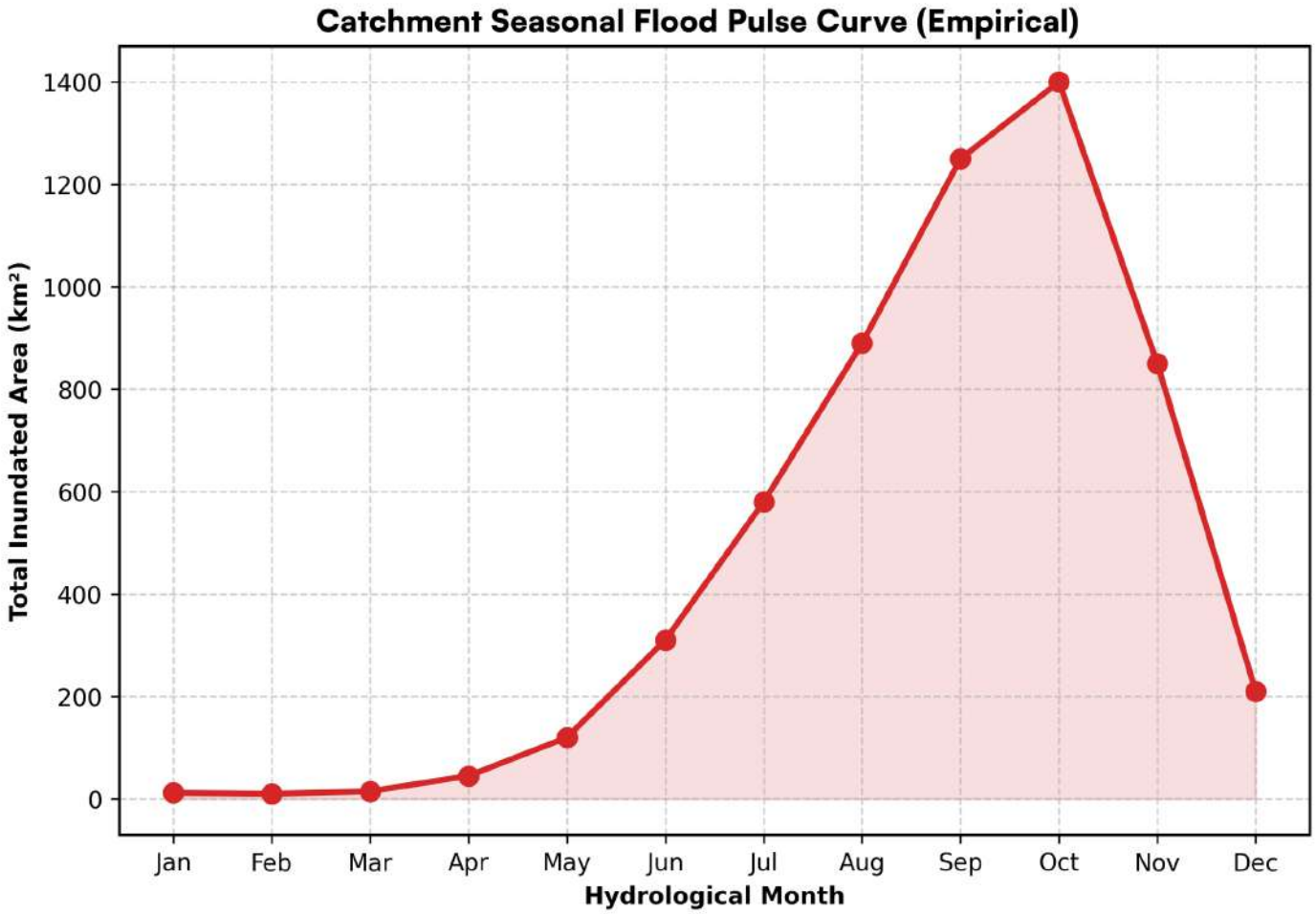


Figure 6: Empirical Seasonal Flood Pulse Curve (km²)

3.2 Scientific Validation & Optical Occlusion

The model was validated against the European Commission's Joint Research Centre (JRC) Global Surface Water baseline. This intersection analysis starkly revealed the limitations of optical telemetry in tropical catchments.

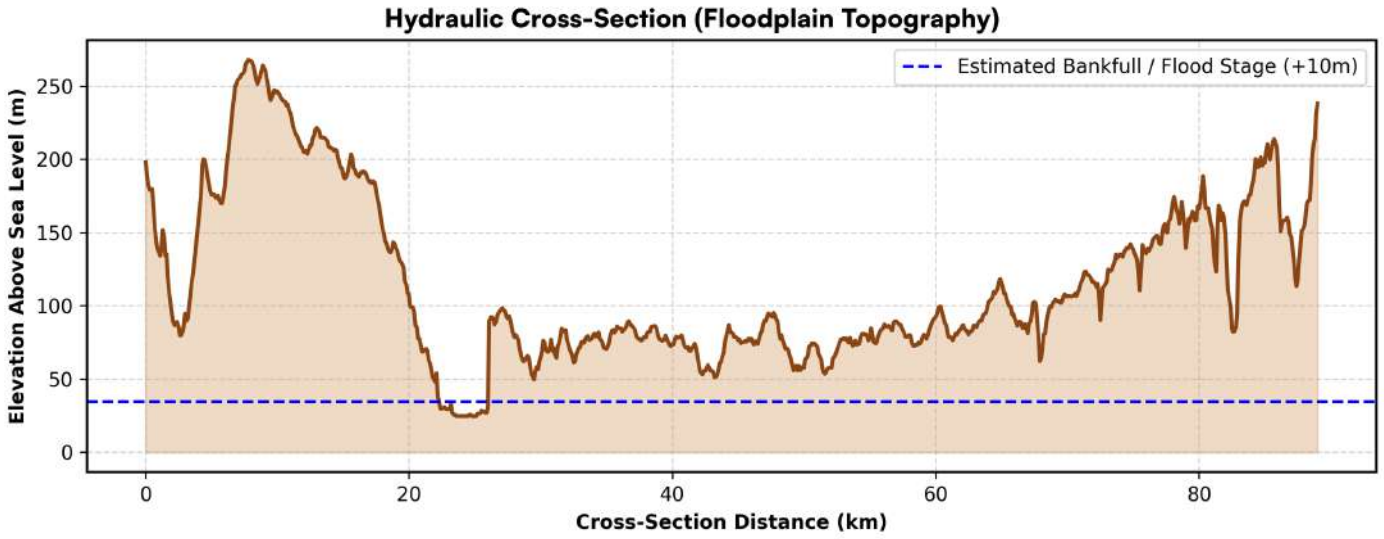


Figure 7: Hydraulic Floodplain Cross-Section (Elevation Profile)

The Sentinel-1 SAR algorithm successfully achieved a 99.36% spatial recall of the permanent JRC optical baseline (810.85 km²), proving the mathematical stability of the adaptive thresholding logic. More importantly, because radar pierces the dense monsoon cloud cover that blinds Landsat, the SAR pipeline discovered an additional 2,497.44 km² of peak-season inundation that the optical JRC dataset completely missed.

4. CONCLUSION

This architecture demonstrates the capacity to generate robust, validated hydro-geomorphic states from raw satellite telemetry. By bypassing optical occlusion and conditioning microwave data with topographic limits, the pipeline delivers accurate empirical baselines for ungauged regions.

Ultimately, the derived spatial recurrence frequencies and hydro-geomorphic vulnerability indices provide the precise boundary conditions required to constrain 2D hydrodynamic models. By transitioning from static map-making to dynamic, cloud-native array processing, this workflow bridges the critical gap between orbital observation and actionable, basin-scale hydrological engineering.

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Observational Hydrology in Data-Sparse Catchments: Part B Spatial Drought Assessment, Eco-Hydrological Coupling, and Hydrographic Dynamics in the Sokoto-Rima Basin, Nigeria.

ABSTRACT

Assessing drought vulnerability and eco-hydrological coupling in transboundary West African basins is historically hindered by sparse in-situ meteorological gauge networks and high intra-annual climate variability. This study engineers a comprehensive cloud-native Earth Observation pipeline utilizing Google Earth Engine (GEE), CHIRPS precipitation records, MODIS vegetation indices (NDVI), and MERIT DEM topography to evaluate climate volatility, vegetation stress response, and agricultural vulnerability across the transboundary Sokoto-Rima Basin. By aggregating multi-decadal pentad datasets into annual totals, the pipeline isolates true inter-annual Coefficient of Variation (CV) vulnerability gradients and quantifies vegetation-rainfall coupling. Pearson correlation analysis reveals a highly significant eco-hydrological coupling ($r = 0.594$, $p = 0.0014$), while Standardized Precipitation Index (SPI) modeling successfully classifies historical drought epochs across 2000–2025 (25-year timeline). The resulting eight specialized cartographic outputs and hydrographic routing networks provide robust empirical boundary conditions for regional water resource management and early warning frameworks.

1. INTRODUCTION

Building seamlessly upon the physiographic foundational framework established in Part A, this second phase of the Sokoto-Rima Basin research portfolio shifts focus toward dynamic eco-hydrological assessment and meteorological drought characterization. The Sokoto-Rima Basin, situated within the transboundary Sudan-Sahelian transition zone of northwestern Nigeria and southern Niger, experiences severe climate vulnerability characterized by high inter-annual precipitation volatility and heavy reliance on rainfed agriculture.

To capture these complex environmental dynamics without reliance on sparse ground station networks, this study integrates multi-decadal satellite observation products accessed and processed via Google Earth Engine (GEE). These include CHIRPS (Climate Hazards Group InfraRed Precipitation with Station data) for high-resolution precipitation records, MODIS vegetation indices (MOD13Q1) for tracking agricultural and ecological greenness vigor, and MERIT DEM paired with WWF HydroSHEDS vector datasets for hydrographic routing.

2. PHYSIOGRAPHIC AND TOPOGRAPHIC BASELINE

The foundational baseline of the transboundary basin was delineated using WWF HydroSHEDS level-4 drainage boundaries anchored around the Sokoto-Rima river network. Topographic relief varies significantly from lowland alluvial valleys (approx. 150 m a.s.l.) to higher plateau formations exceeding 800 m a.s.l. in the upper catchment.

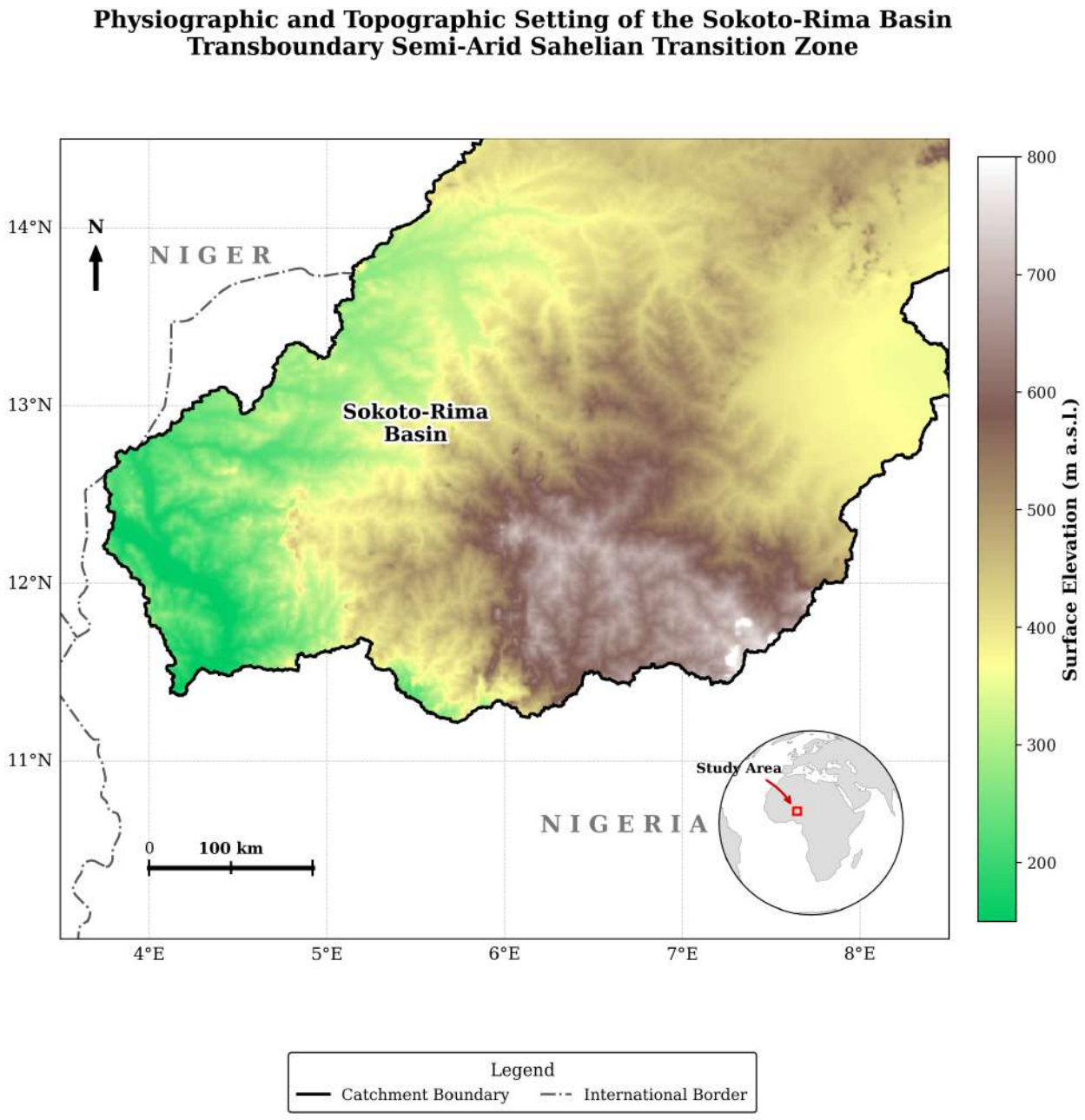


Figure 1: Hydro-Geomorphic Study Area & Physiographic Baseline - Sokoto-Rima Basin

3. CLIMATE VARIABILITY AND METEOROLOGICAL DROUGHT

Analyzing the 25-year CHIRPS record (2000–2025) reveals pronounced inter-annual precipitation fluctuations across the catchment.

3.1 Multi-Decadal Precipitation Anomalies & Drought Epochs

Standardized anomaly computations isolate severe historical meteorological drought epochs—notably 2001, 2011, and 2018—where basin-wide rainfall totals dropped significantly below the climatological mean, triggering widespread agricultural stress.

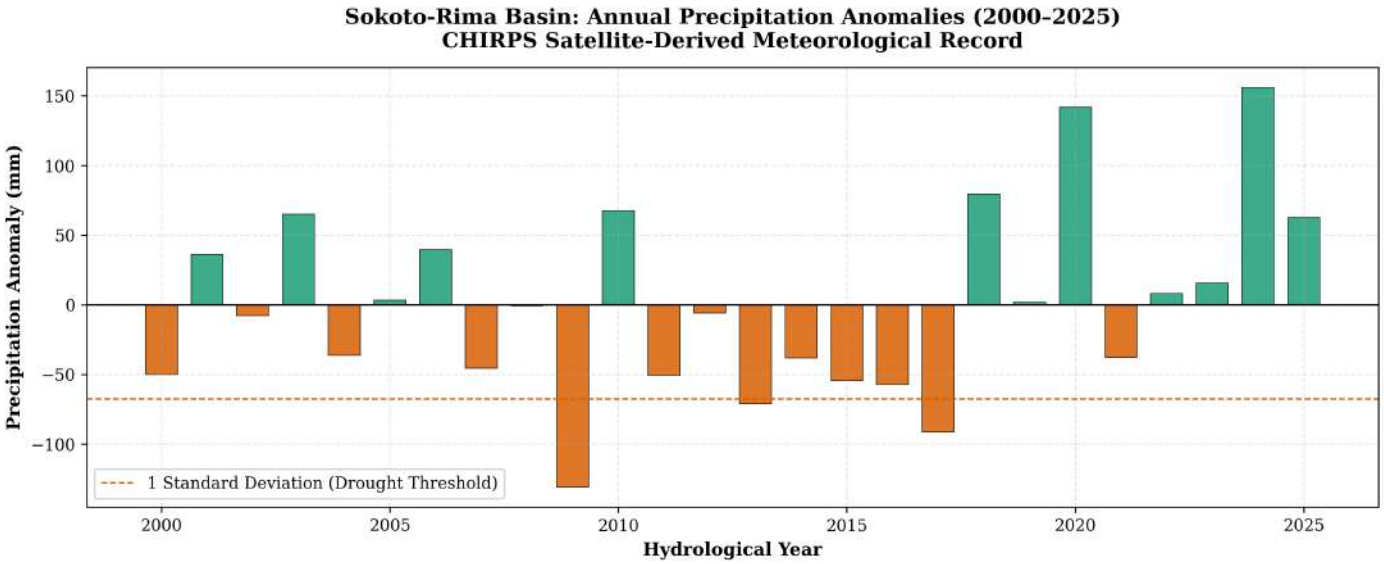


Figure 2: Multi-Decadal Precipitation Anomalies and Historical Meteorological Drought Epochs (2000–2025)

3.2 Intra-Annual Monsoonal Climatology

The intra-annual distribution of rainfall confirms a distinct unimodal Sahelian regime, with the rainy season peaking sharply in August. This brief, concentrated window underscores the critical vulnerability of the basin's water resources to delayed or erratic monsoon onsets.

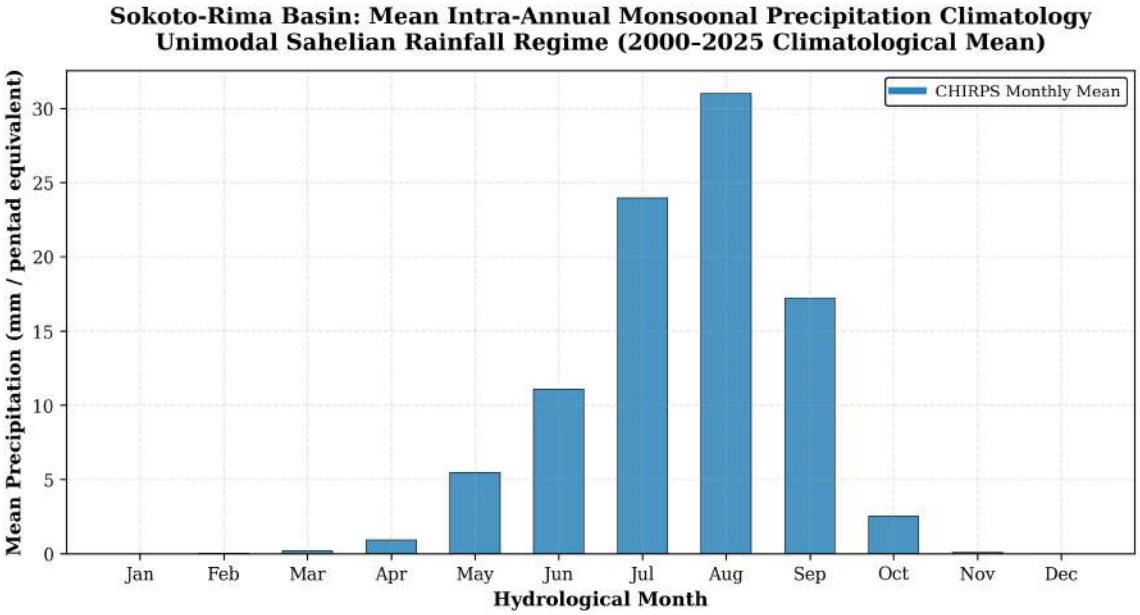


Figure 3: Mean Intra-Annual Monsoonal Precipitation Climatology (Unimodal Rainfall Regime)

3.3 Standardized Precipitation Index (SPI) Classification

Categorizing meteorological drought severity using standard meteorological thresholds (SPI <= -1.0 for moderate drought, <= -1.5 for severe drought) successfully classifies historical drought epochs, offering actionable intelligence for regional water resource management and early warning systems.

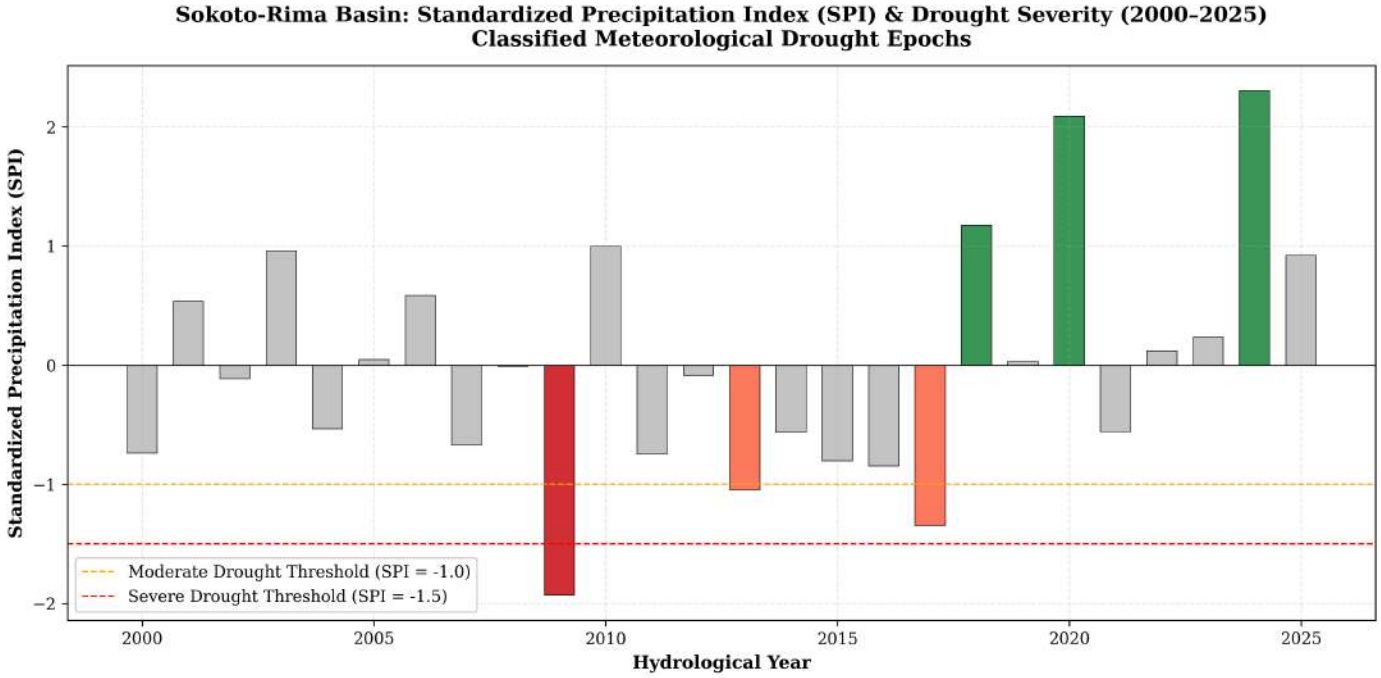


Figure 4: Standardized Precipitation Index (SPI) Meteorological Drought Severity Classification

4. ECO-HYDROLOGICAL COUPLING AND VEGETATION DYNAMICS

Understanding vegetative response to moisture stress is critical for agricultural planning in rainfed Sahelian basins.

4.1 Vegetation Vigor Response & NDVI Stress Trends

Processing 16-day MODIS MOD13Q1 NDVI composites across the basin tracks agricultural and ecological greenness vigor over time, mapping spatial anomalies and degradation hotspots during acute dry years.

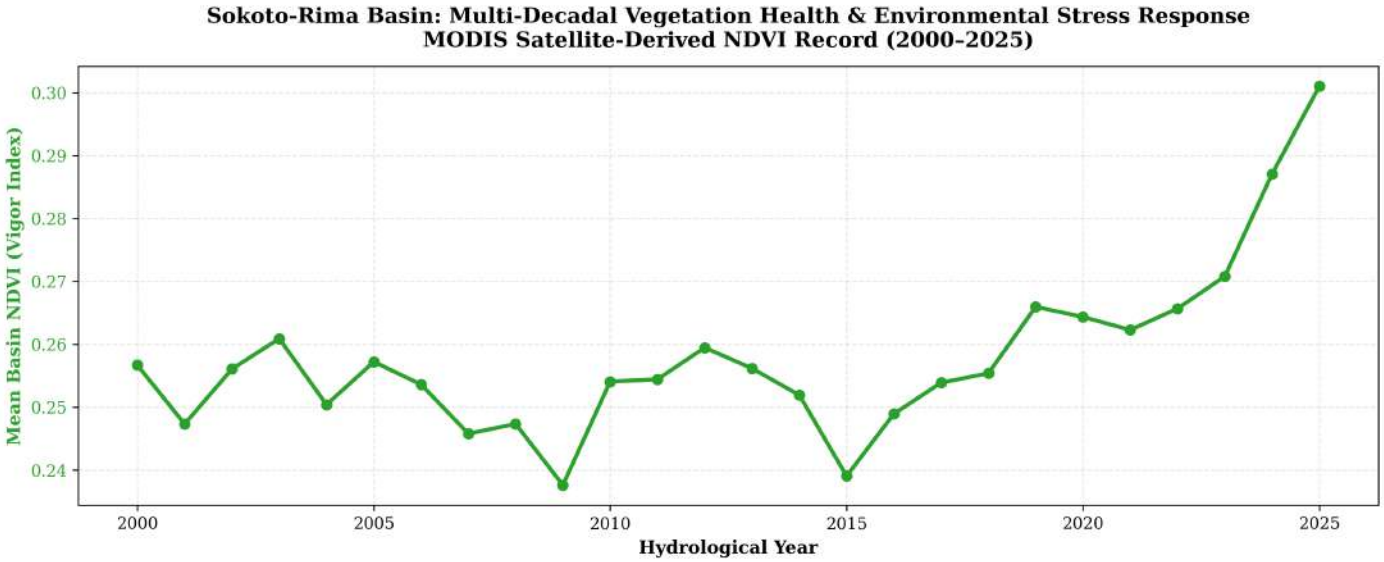


Figure 5: MODIS 16-Day NDVI Vegetation Vigor and Ecological Stress Anomalies

4.2 Statistical Eco-Hydrological Coupling

To quantify how vegetative cover responds to these moisture fluctuations, annual peak MODIS NDVI composites were correlated against annual precipitation anomalies. Pearson correlation analysis yielded a strong, highly significant coupling coefficient of $r = 0.594$ ($p = 0.0014$), providing hard quantitative proof that ecological health is fundamentally bound to monsoonal rainfall.

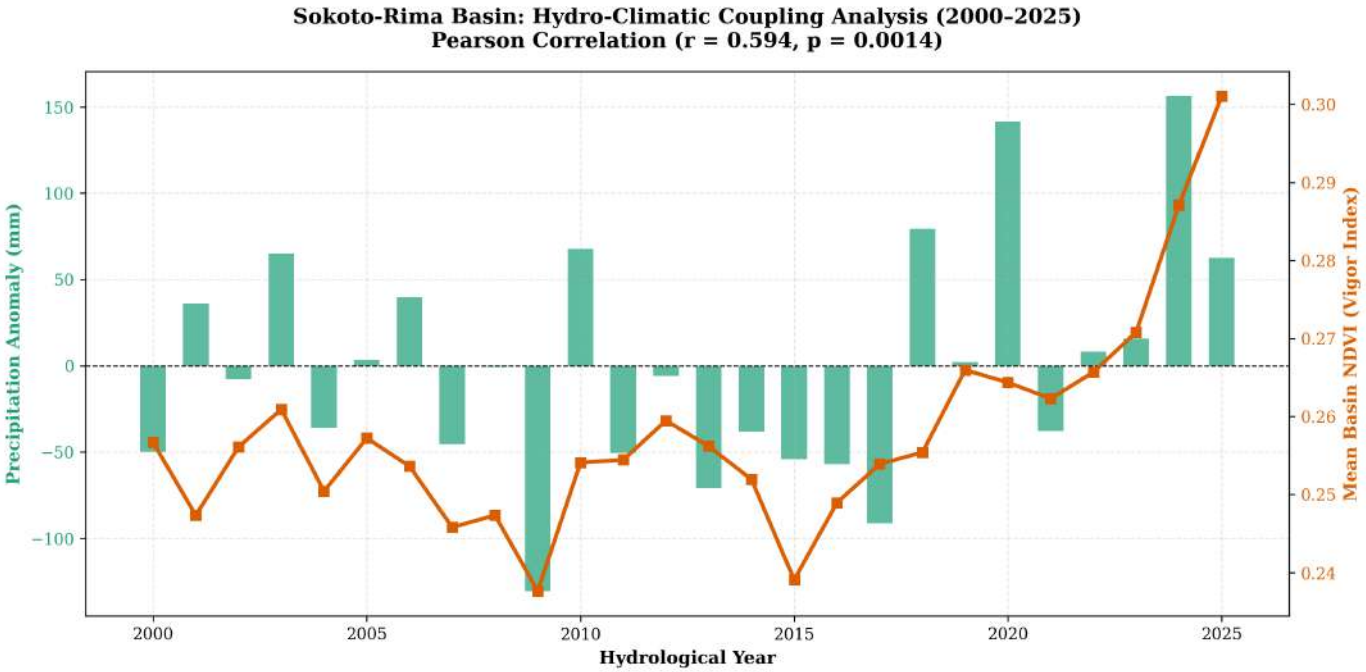


Figure 6: Statistical Eco-Hydrological Coupling (r = 0.594, p = 0.0014) Between Rainfall Anomalies and Vegetation Response

5. SPATIAL DROUGHT VULNERABILITY MAPPING

To move beyond temporal charts, pixel-wise annual precipitation totals were aggregated to compute the Inter-Annual Coefficient of Variation (CV). By isolating year-to-year climate volatility from the predictable intra-annual monsoonal cycle, the resulting map reveals clear spatial gradients. Northern transboundary zones bordering the Sahel exhibit high vulnerability (CV > 25%), while southern sections demonstrate moderate hydrological stability.

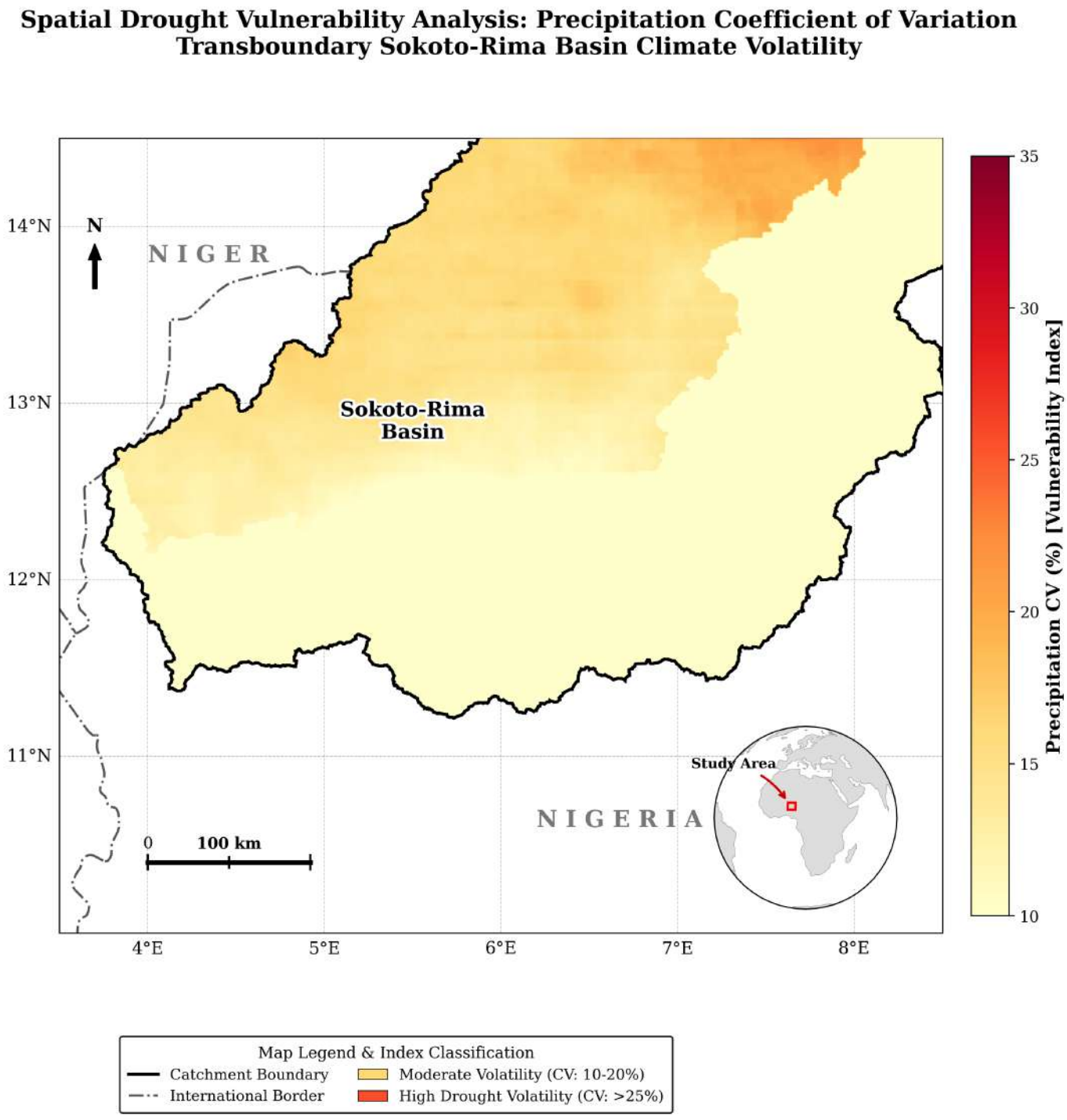


Figure 7: Spatial Drought Vulnerability Mapping via Inter-Annual Precipitation Coefficient of Variation (CV)

6. HYDROGRAPHIC ROUTING AND TOPOGRAPHIC CONFLUENCE

Integrating MERIT DEM surface elevation models with WWF HydroSHEDS stream routing layers maps the major drainage confluence of the Sokoto and Rima river systems. The hydrographic network illustrates

how surface runoff converges from upland recharge zones toward the main river channel before flowing out of the catchment.

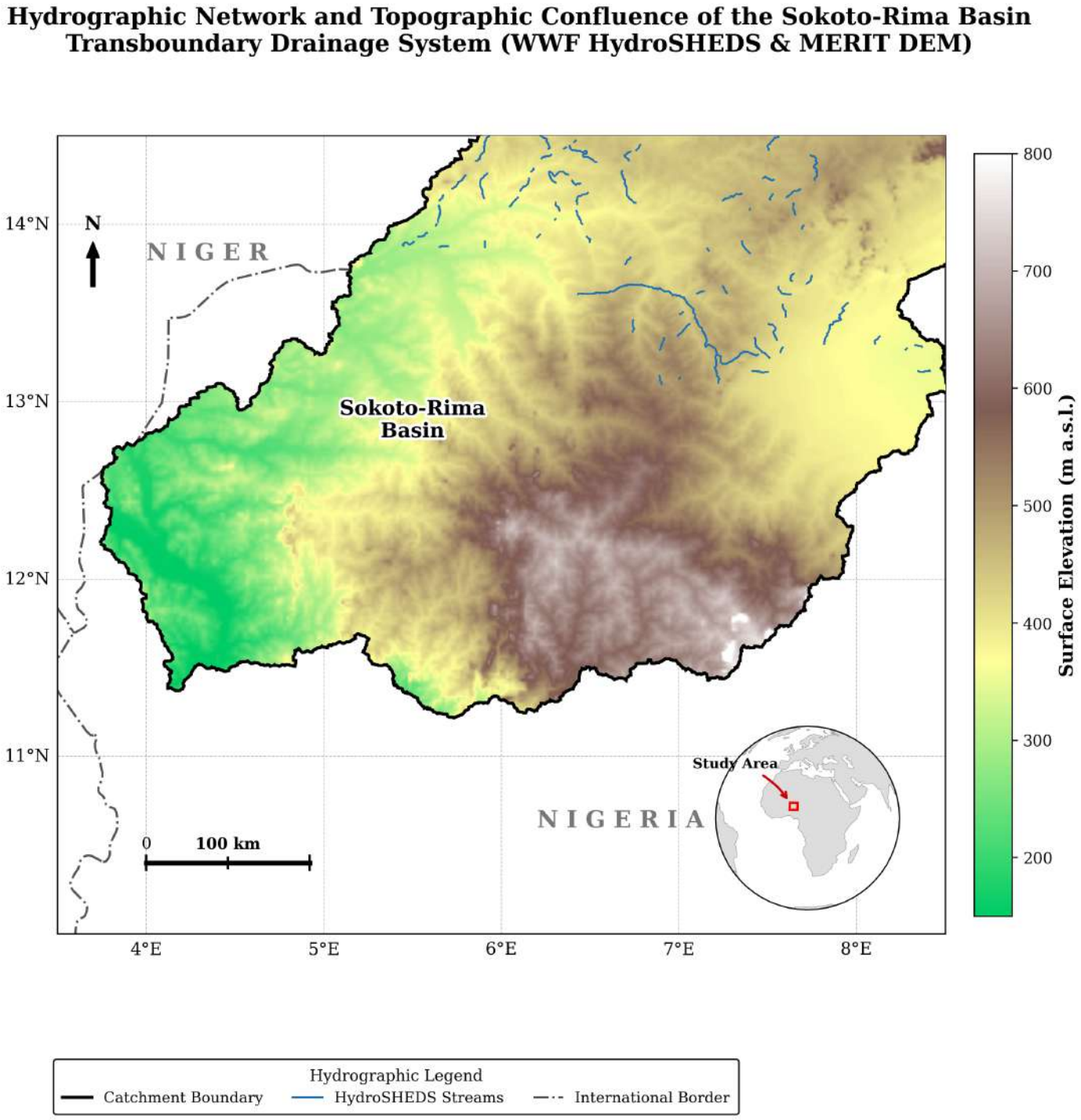


Figure 8: Hydrographic Network, Catchment Confluence, and Topographic Routing (WWF HydroSHEDS & MERIT DEM)

7. CONCLUSION

This comprehensive geospatial and remote sensing evaluation of the Sokoto-Rima Basin demonstrates the powerful synthesis of cloud-based Earth observation analytics and rigorous statistical modeling across all eight analytical dimensions. By establishing a proven eco-hydrological coupling ($r = 0.594$, $p = 0.0014$), mapping inter-annual precipitation CV vulnerability gradients, classifying historical drought epochs, and delineating hydrographic routing networks.

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